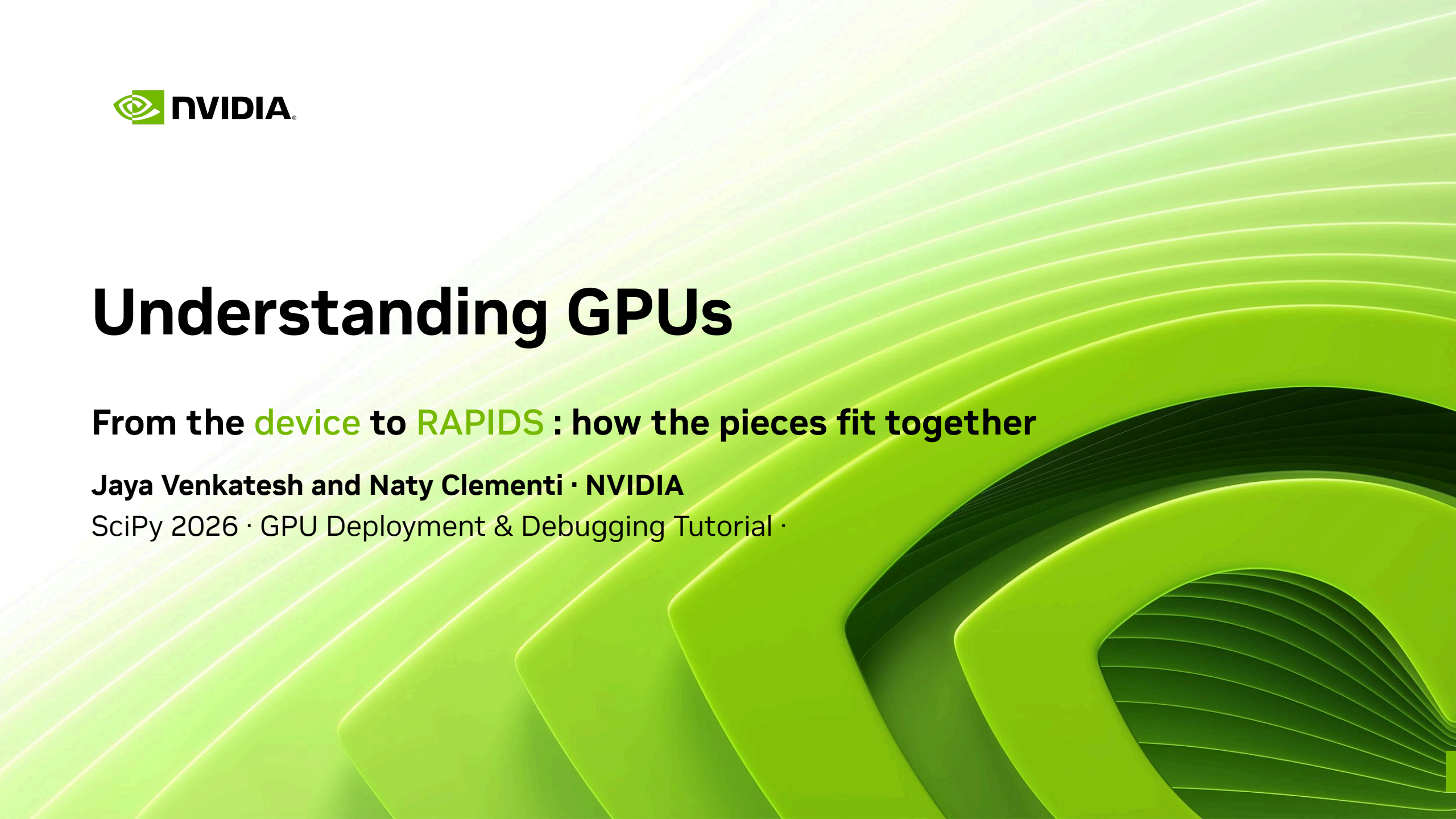


# Understanding GPUs

## From the device to RAPIDS : how the pieces fit together

**Jaya Venkatesh and Naty Clementi · NVIDIA**

SciPy 2026 · GPU Deployment & Debugging Tutorial ·

# Why bring data science to the GPU?

**Bigger data**
Datasets keep outgrowing what's comfortable on a CPU

**Parallel by nature**
Filtering, aggregating, and math over millions of rows are independent

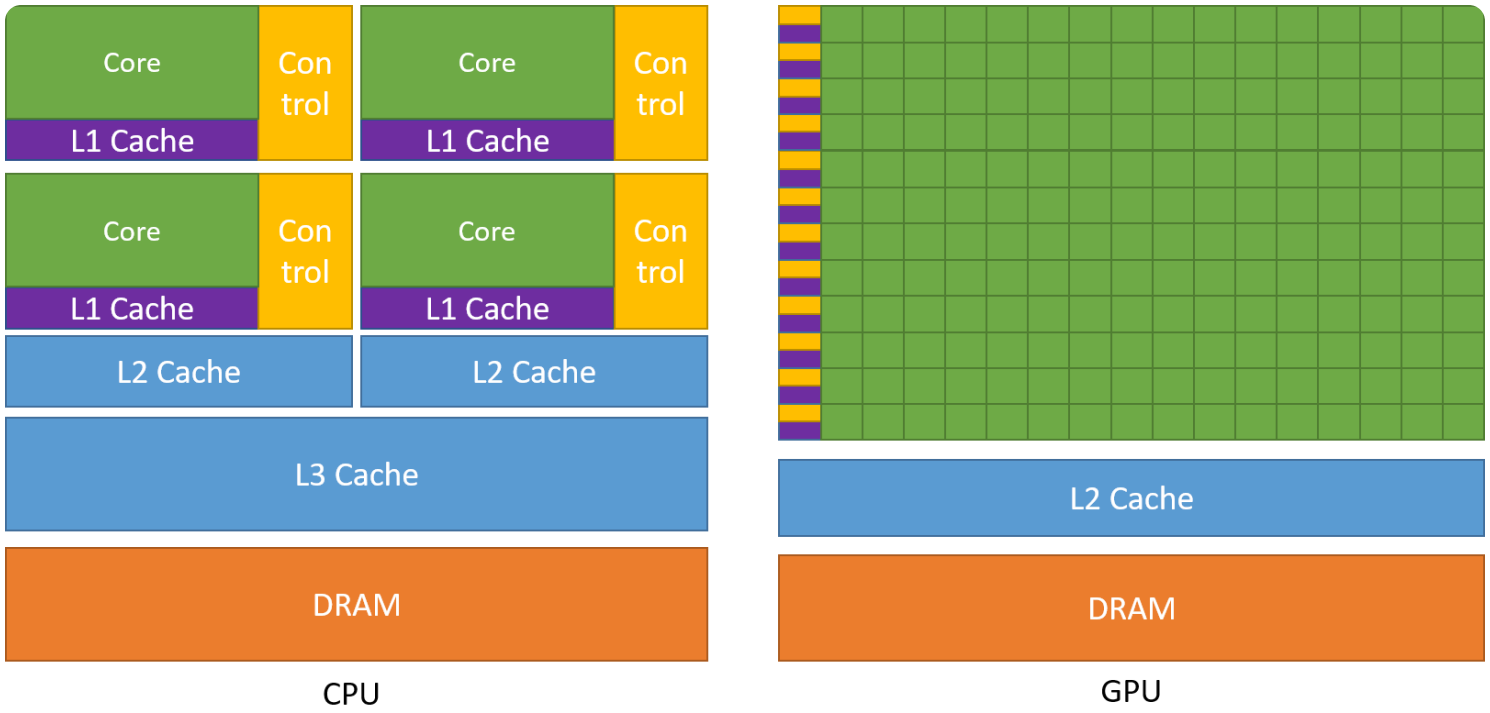
**Familiar APIs**
You can get there without leaving Python

But how does a GPU actually **help you accelerate your code?**

# 1 · The GPU vs the CPU

Both process data, just with a different philosophy

# A CPU and a GPU are built for different jobs




Source: NVIDIA CUDA C++ Programming Guide

**CPU**: handles **many different tasks**, finishing **each one fast** → *low latency*.

**GPU**: runs the **same task across lots of data** all at once → *high throughput*.



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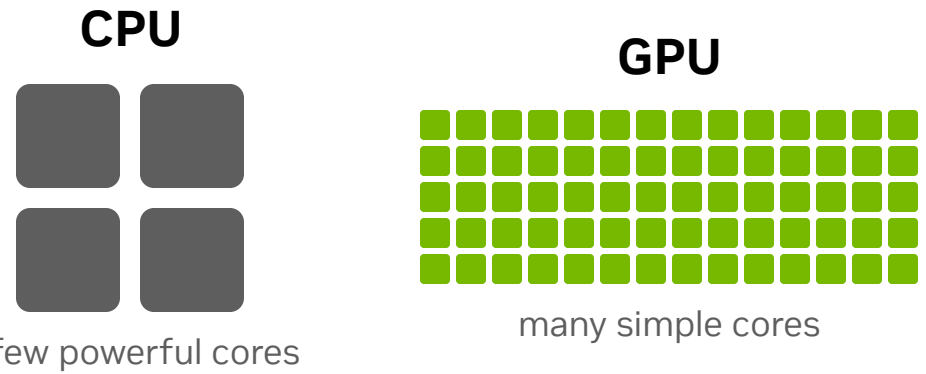
# A few specialists vs many simple workers



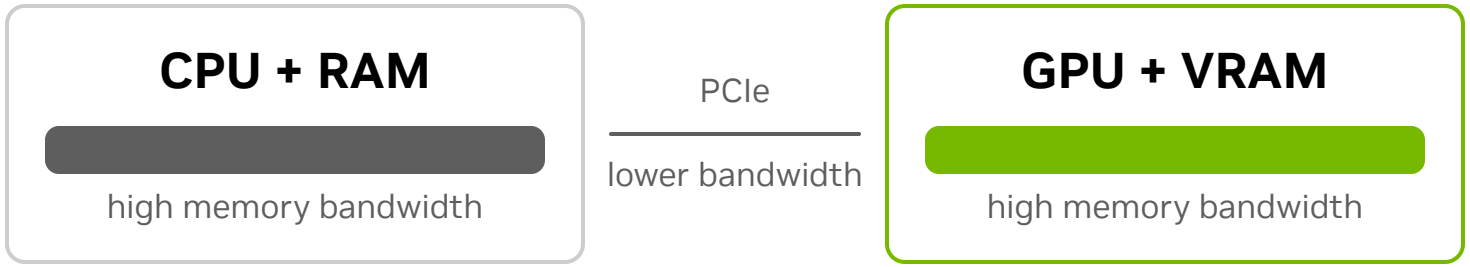

- CPU cores are **few but powerful**, great at complex, branchy logic
- GPU cores are **many but simple**, built to all do the **same operation** on different data, together
- The more your work splits into **independent pieces**, the more **using the GPU pays off**

# Two machines, two separate memories


- The GPU is a **separate device** with its **own memory**: it **cannot** read data sitting in system RAM
- Before the GPU can work on your data, that data must be **copied into the GPU's memory**

So step one of "run it on the GPU" is always: **get the data there.**

# The slow part is getting data *across*



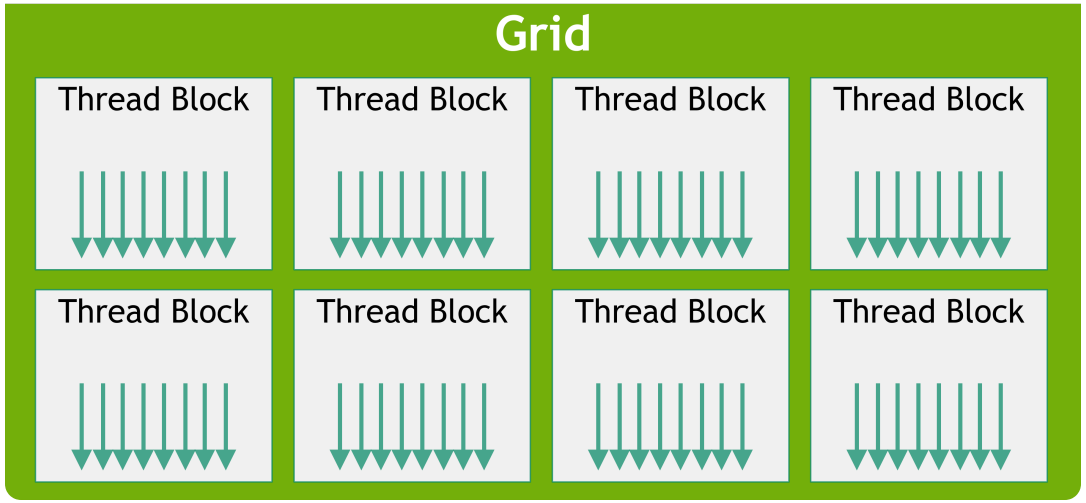

- **Inside** each device, **memory bandwidth is high**: RAM and VRAM are built to feed their cores fast
- The **interconnect between them (PCIe)** carries data at **far lower bandwidth**
- The expensive part often isn't the *computing*; it's the **host-to-device transfer** across PCIe

# 2 · How a GPU processes data

Many small tasks at once, not one big task in order

# Parallel work: threads, blocks, grids

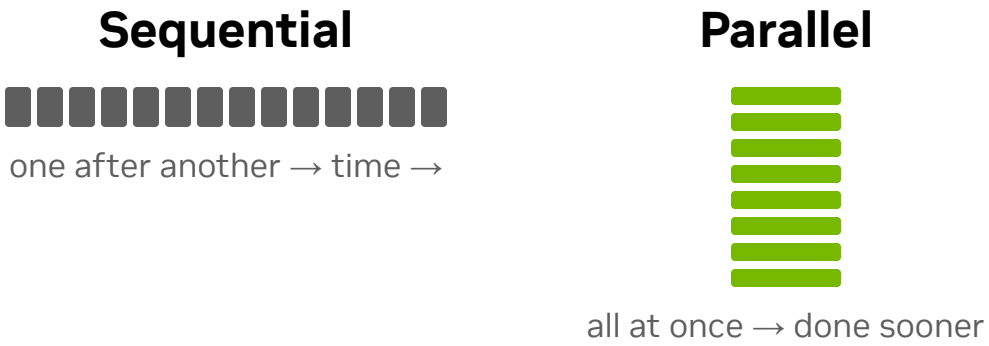
The GPU runs **one operation across thousands of elements at once**, one **thread** per item.




Source: NVIDIA CUDA C++ Programming Guide

- Threads are grouped into **blocks**; blocks together form a **grid**
- You describe the work *once*; the GPU runs it across the **whole grid in parallel**
- Perfect for **independent** work, not optimal for **sequential**, step-by-step logic

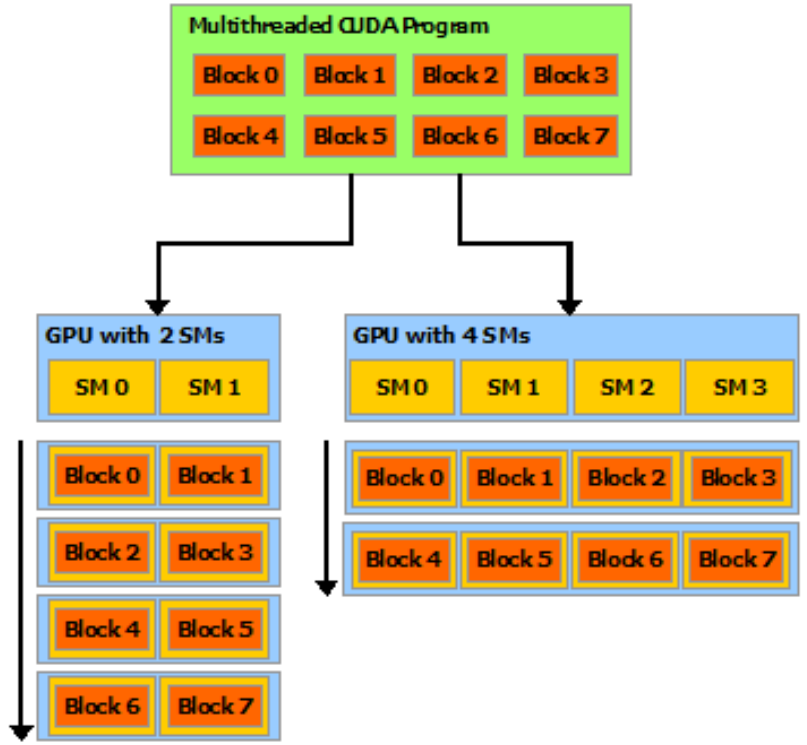
# Why GPU is better for parallel than sequential




- When work items are **independent**, their order doesn't matter: item B doesn't need item A's result
- A GPU runs a **huge number of independent items at the same time**
- Step-by-step work that must run **in order** can't use that; this isn't where a GPU shines

# SMs and warps: where the work runs

- A GPU is made of **Streaming Multiprocessors (SMs)**, its parallel engines
- Your **blocks are distributed across the SMs**; more SMs → more work at once, **automatically**
- Inside an SM, threads run in lock-step groups of **32, called a warp** *(SIMT: Single Instruction, Multiple Threads)*




Source: NVIDIA CUDA C++ Programming Guide

# Compute-bound vs IO-bound

**✓ Compute-bound: GPU shines**

- A matrix multiply, training a model, transforming millions of rows
- **Lots of math** → the cores stay saturated

**✗ IO-bound: GPU waits**

- Reading a file off disk, waiting on a network call
- The bottleneck is **waiting**, not math

Adding compute units only helps when **computation** is the bottleneck. If you're waiting on data, more cores don't make the wait shorter.

# So what actually fits a GPU?

**✓ Parallel**
The same operation over millions of elements

**✗ Sequential**
Each step depends on the previous one

**✓ Compute-bound**
Lots of math per byte of data

**✗ IO-bound**
Time spent waiting on disk / network

Takeaway: a GPU shines on **big, math-heavy, parallel** work, so the goal is to match your workload to its strengths.

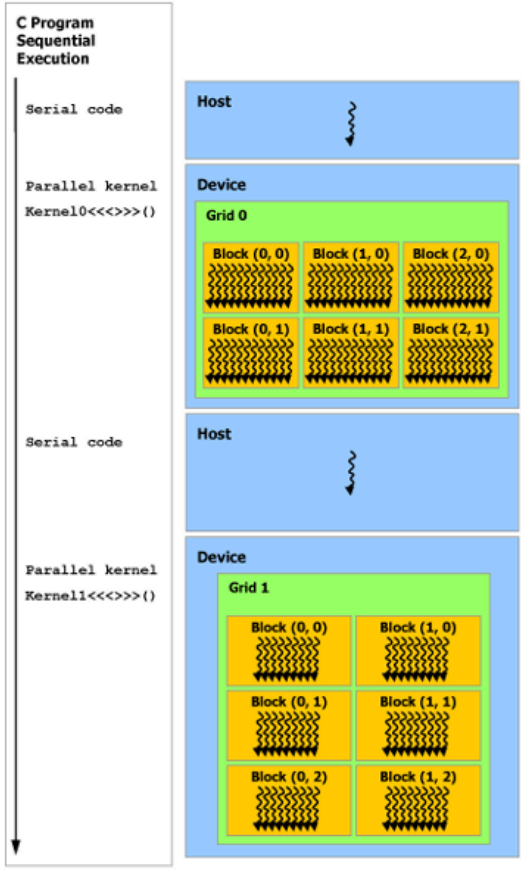
# 3 · CUDA

The bridge that lets software use the GPU

# What is CUDA, and why C?

**CUDA** is NVIDIA's parallel-computing **platform + programming model**, built on **C/C++** for its low-level speed and control over the hardware.

- You write GPU code as **kernels** in **CUDA C/C++**: standard C/C++ plus a few extensions, compiled by `nvcc`
- The **host** (CPU) runs serial code and **launches kernels** on the **device** (GPU)
- Serial host code and parallel kernels **alternate** (see diagram →)




Source: NVIDIA CUDA C++ Programming Guide



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# A CUDA kernel is just C, with a twist

```
// __global__ marks a function that runs ON the GPU, once per thread
__global__ void add(float *a, float *b, float *c) {
    int i = blockIdx.x * blockDim.x + threadIdx.x;  // which element am I?
    c[i] = a[i] + b[i];                             // one thread, one element
}

// The host launches it across a grid:  <<< blocks, threads-per-block >>>
add<<<blocks, threads>>>(a, b, c);   // every element added in parallel
```

The `__global__` keyword and the `<<< >>>` launch syntax are the **C extensions** CUDA adds. A sum, a sort, a join all become kernels like this, but **you'll rarely write one**: the layers above ship them ready-made.

# 4 · CUDA Python & the CUDA Toolkit

The building blocks above raw CUDA

# The CUDA Toolkit: the foundation libraries

Everything above the driver is **built on the Toolkit**. It provides:

**Compiler & runtime**
nvcc, libcudart, headers

**Math libraries**
cuBLAS, cuFFT, cuSPARSE, cuSOLVER, cuRAND

**Communication**
NCCL, NVSHMEM for multi-GPU

**Why it's central**
cuDF, cuML & CuPy don't reinvent math, they call these

These hand-tuned libraries are the **ready-made kernels**: the sorts, joins, and math behind your data work come straight from here.

# CUDA Python: pick the right door

You don't have to write C++ to use CUDA. Match the door to the need:

**CuPy**
"I have NumPy code" → swap in GPU arrays

**Numba (CUDA)**
"I need a custom kernel" → write it in Python

**cuda.core / cuda.bindings**
"I'm building a library / need low-level control"

All three reach the **same CUDA underneath**. RAPIDS sits on top of them, so most of the time you won't pick a door at all.

# 5 · RAPIDS & CUDA-X

Predefined kernels for data science

# CUDA-X & RAPIDS: kernels you don't have to write

**CUDA-X**

NVIDIA's library collection on top of CUDA: math, deep learning (cuDNN, TensorRT), comms, data science

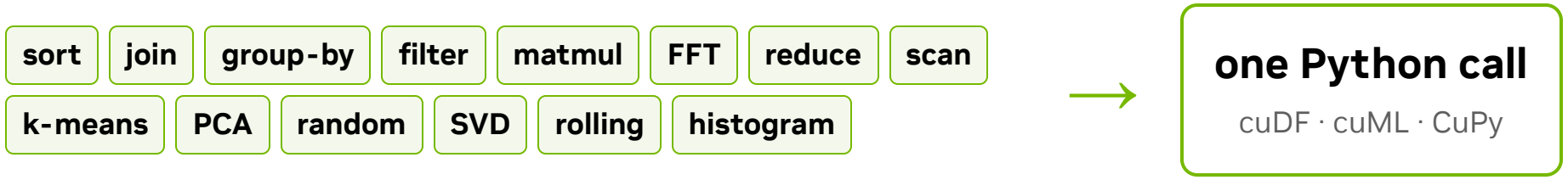
**RAPIDS**

The **data-science slice** of CUDA-X, open-source, with familiar Python APIs

RAPIDS packages thousands of **optimized GPU kernels** behind APIs you already know:

| You already know | RAPIDS gives you | for |
|---|---|---|
| `pandas` · `scikit-learn` · `NumPy` · `NetworkX` | **cuDF** · **cuML** · **CuPy** · **cuGraph** | dataframes · ML · arrays · graphs |

# You stand on thousands of tuned kernels




- Each tile is a **GPU kernel** that experts wrote and tuned over years
- RAPIDS bundles them behind the **APIs you already know**
- You write familiar Python; **RAPIDS runs the CUDA for you**

*How much* faster on real data? We'll explore that in this tutorial

# Putting it all together

**Your code & notebooks**
pandas, scikit-learn, NumPy, your scripts

**RAPIDS**
cuDF · cuML · cuGraph, the data-science slice of CUDA-X

**CUDA Python**
cuda.core · cuda.bindings · Numba · CuPy

**CUDA-X + CUDA Toolkit**
cuBLAS · cuSPARSE · NCCL · libcudart · nvcc

**NVIDIA driver**
libcuda, the system layer that talks to the hardware

**GPU hardware**
SMs · warps · thousands of cores · VRAM

Each layer builds on the one below it; your Python sits at the very top.

# The mental model for today

- A GPU is **many simple cores + its own fast memory** → built for **parallel, math-heavy** work
- Getting data **onto** the GPU is the slow part → **keep it there**
- **CUDA** (C-based) exposes that power, but you rarely touch it
- **CUDA Python** and the **CUDA Toolkit** are the building blocks
- **RAPIDS / CUDA-X** hand you **ready-made kernels**, so familiar Python code just runs on the GPU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Next: let's get on a GPU